

ON DIMER COVERINGS OF RECTANGLES OF FIXED WIDTH

Richard P. STANLEY*

Department of Mathematics, Massachusetts Institute of Technology, Cambridge, MA 02139, USA

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For fixed k let A_n denote the number of dimer coverings of a $k \times n$ rectangle. Various properties of the generating function $\sum A_n x^n$ are obtained, in particular answering questions of Klarner and Pollack and of Hock and McQuistan. An explicit expression for the molecular freedom for dimers on a saturated $k \times n$ lattice space is also obtained. The results are consequences of the explicit formula for A_n obtained by Kasteleyn and by Temperley and Fisher.

Let k be a fixed positive integer, and let $A_n = A_{n,k}$ denote the number of ways to tile a $k \times n$ rectangle with $nk/2$ dimers (or dominoes). (Of course $A_n = 0$ if nk is odd.) Form the generating function

$$F_k(x) = \sum_{n \geq 0} A_n x^n.$$

It is well known (e.g., [5]) that $F_k(x)$ represents a rational function, say $F_k(x) = P_k(x)/Q_k(x)$ with P_k, Q_k polynomials with integer coefficients, and $Q_k(0) = 1$. We do not assume that $F_k(x)$ is reduced to lowest terms. If

$$Q_k(x) = 1 - \alpha_1 x - \cdots - \alpha_q x^q,$$

then it follows that

$$A_{n+q} = \alpha_1 A_{n+q-1} + \cdots + \alpha_q A_n \quad (1)$$

for all n sufficiently large (and for all $n \geq 0$ if and only if $\deg P_k < \deg Q_k$; we will show below that $\deg Q_k - \deg P_k = 2$). For the basic facts concerning rational generating functions, see [8]. The largest root of the polynomial $x^q Q_k(1/x)$, when $F_k(x)$ is reduced to lowest terms, is denoted by μ_k ; and the number $\lambda_k = \mu_k^{2/k}$ is called the *molecular freedom* for dimers on a saturated $k \times n$ lattice space.

Recently Klarner and Pollack [5] computed $P_k(x)$ and $Q_k(x)$ for $1 \leq k \leq 8$, while Hock and McQuistan [3] computed $Q_k(x)$ for $1 \leq k \leq 10$. They also computed numerically the values of μ_k for $1 \leq k \leq 8$ and $1 \leq k \leq 10$, respectively. Both papers raised various questions about the properties of $P_k(x)$ and $Q_k(x)$. Here we will

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answer these and other questions and will give an explicit formula for μ_k .

Our results are direct consequences of Kasteleyn's formula for A_n (also obtained by Temperley and Fisher [9] and later by Lieb [6]), which Kasteleyn shows [4, eqn. (15)] can be written in the form

$$A_n = \begin{cases} \prod_{j=1}^{\lfloor k/2 \rfloor} \frac{c_j^{n+1} - \bar{c}_j^{n+1}}{2b_j}, & nk \text{ even,} \\ 0, & nk \text{ odd} \end{cases} \quad (2)$$

where

$$\begin{aligned} c_j &= \cos \frac{j\pi}{k+1} + \left(1 + \cos^2 \frac{j\pi}{k+1}\right)^{1/2}, \\ \bar{c}_j &= \cos \frac{j\pi}{k+1} - \left(1 + \cos^2 \frac{j\pi}{k+1}\right)^{1/2}, \\ b_j &= \left(1 + \cos^2 \frac{j\pi}{k+1}\right)^{1/2}. \end{aligned}$$

Note that $c_j \bar{c}_j = -1$.

Write $l = \lfloor k/2 \rfloor$, and let S be any subset of $\{1, \dots, l\}$ and $\bar{S} = \{1, \dots, l\} - S$. Define

$$c_S = \left(\prod_{j \in S} c_j \right) \left(\prod_{j \in \bar{S}} \bar{c}_j \right).$$

Then (2) shows that

$$A_n = \left[\prod_{j=1}^l (2b_j)^{-1} \right] \sum_S (-1)^{|\bar{S}|} c_S^{n+1}, \quad (3)$$

provided nk is even, where S ranges over all subsets of $\{1, \dots, l\}$.

Lemma. *We have*

$$\prod_{j=1}^l b_j^2 = d_k 2^{-k},$$

where $d_0 = 1$, $d_1 = 2$, $d_k = 2d_{k-1} + d_{k-2}$. Explicitly,

$$d_k = \frac{(1 + \sqrt{2})^{k+1} - (1 - \sqrt{2})^{k+1}}{2\sqrt{2}}. \quad (4)$$

Proof. When k is even, equation (4) is the case $u = 1$ of a formula of Kasteleyn [4, eqn. (14)], and when k is odd, Kasteleyn's proof is also valid. The recurrence $d_k = 2d_{k-1} + d_{k-2}$ follows from (4) since $1 \pm \sqrt{2}$ are roots of the polynomial $x^2 - 2x - 1$.

One can also view this lemma (as well as [4, eqn. (14)]) as a standard result on the Chebyshev polynomial

$$U_k(x) = 2^k \prod_{j=1}^k \left(x - \cos \frac{j\pi}{k+1} \right),$$

after observing that

$$i^k 2^k \prod_{j=1}^l b_j^2 = U_k(i), \quad \text{where } i^2 = -1.$$

Theorem. (a) The polynomial $Q_k(x)$ can be taken to be

$$Q_k(x) = \begin{cases} \prod_S (1 - c_S x), & k \text{ even} \\ \prod_S (1 - c_S^2 x^2), & k \text{ odd}, \end{cases} \quad (5)$$

where S ranges over all subsets of $\{1, \dots, l\}$. Hence A_n satisfies a linear recurrence (1) (which by (d) below will be valid for all $n \geq 0$) of degree $q_k = \deg Q_k = 2^{\lfloor (k+1)/2 \rfloor}$. Moreover, all the roots of $Q_k(x)$ are real and nonzero, and exactly half the roots are positive.

(b) The largest reciprocal root of $Q_k(x)$ is

$$\mu_k = \prod_{j=1}^l c_j, \quad (6)$$

which occurs with multiplicity one and which is not a reciprocal root of $P_k(x) = Q_k(x)F_k(x)$. The molecular freedom $\lambda_k = \mu_k^{2/k}$ satisfies

$$\lambda = \lim_{k \rightarrow \infty} \lambda_k = e^{2G/\pi} = 1.79162\dots, \quad (7)$$

where $G = \sum_{s \geq 0} (-1)^s (2s+1)^{-1}$ is Catalan's constant.

(c) Asymptotically we have

$$A_n \sim a_k \mu_k^{n+1}, \quad \text{as } n \rightarrow \infty \text{ with } nk \text{ even}, \quad (8)$$

where

$$a_k = 2^\delta d_k, \quad (9)$$

where d_k is given by (4) and where $\delta = 0$ if k is even and $\delta = \frac{1}{2}$ if k is odd. Moreover,

$$\lim_{k \rightarrow \infty} a_k^{2/k} = 1/(\sqrt{2} + 1) = \sqrt{2} - 1.$$

(d) $P_k(x)$ has degree $p_k = 2^{\lfloor (k+1)/2 \rfloor} - 2 = q_k - 2$. Hence A_n satisfies (1) for all $n \geq 0$.

(e) If $k > 1$, then $P_k(x) = -x^{p_k} P_k(1/x)$. If k is odd or divisible by 4, then $Q_k(x) = x^{q_k} Q_k(1/x)$. If $k \equiv 2 \pmod{4}$, then $Q_k(x) = -x^{q_k} Q_k(1/x)$. If k is odd, then $P_k(x) = P_k(-x)$ and $Q_k(-x) = Q_k(x)$. (The statements about $Q_k(x)$ are equivalent to property (d) of the roots observed by Hock and McQuistan [3, p. 104] for $k \leq 10$.)

(f) For k odd write

$$\begin{aligned} Q_k(x) &= \beta_0 - \beta_1 x^2 + \beta_2 x^4 - \dots + \beta_r x^{2r} \\ &= \gamma_0 - \binom{r}{1} \gamma_1 x^2 + \binom{r}{2} \gamma_2 x^4 - \dots + \gamma_r x^{2r}, \end{aligned}$$

where $r=2^l$. Then the numbers γ_i are positive and log-concave (i.e., $\gamma_i^2 \geq \gamma_{i-1}\gamma_{i+1}$). Thus they are also unimodal (i.e., increase monotonically to a maximum, and then decrease monotonically). (This implies that the β_i 's are also positive, log-concave, and unimodal.)

(g) Define

$$T_k(x) = \begin{cases} \prod_{|S| \text{ even}} (1 - c_S x), & k \text{ even}, \\ \prod_{|S| \text{ even}} (1 - c_S^2 x^2), & k \text{ odd}, \end{cases}$$

$$\bar{T}_k(x) = \begin{cases} \prod_{|S| \text{ odd}} (1 - c_S x), & k \text{ even}, \\ \prod_{|S| \text{ odd}} (1 - c_S^2 x^2), & k \text{ odd}, \end{cases}$$

so that $Q_k(x) = T_k(x)\bar{T}_k(x)$. Then the coefficients of $T_k(x)$ and $\bar{T}_k(x)$ lie in the field $\mathbb{Q}(d_k^{1/2})$, where d_k is given by the Lemma, and if $d_k^{1/2} \notin \mathbb{Q}$, then the coefficients of any monomial x^j in $T_k(x)$ and $\bar{T}_k(x)$ are conjugate in $\mathbb{Q}(d_k^{1/2})$. If $d_k^{1/2} \in \mathbb{Q}$, then $T_k(x)$ and $\bar{T}_k(x)$ have rational coefficients (so $Q_k(x)$ is reducible over $\mathbb{Q}$). (J. Lagarias has shown me a proof that d_k is a square if and only if $k=0$ or $k=6$). When $k=6$ we have

$$T_6(x) = (1-x)(1-6x+5x^2-x^3),$$

$$\bar{T}_6(x) = (1+x)(1+5x+6x^2+x^3).$$

(The fact that ± 1 are roots of $Q_6(x)$ is equivalent to the surprising identity $c_1 = c_2 c_3$ for $k=6$.) Moreover, when k is even,

$$P_k(x) = d_k^{-1/2} (T_k(x)\bar{T}'_k(x) - T'_k(x)\bar{T}_k(x)).$$

Proof. (a) From (2) it follows that $F_k(x) = A_k(x)/B_k(x)$, where $B_k(x) = \prod_S (1 - c_S x)$ and where $A_k(x)$ is a polynomial. Hence to prove (5) it suffices to show that the coefficients of $C_k(x)$ are integers where

$$C_k(x) = \begin{cases} B_k(x), & k \text{ even}, \\ B_k(x)B_k(-x), & k \text{ odd}. \end{cases}$$

Equivalently, if σ is an automorphism of the splitting field of the field $L = \mathbb{Q}(c_S \mid S \subseteq \{1, \dots, l\})$ (actually, L is Galois extension of $\mathbb{Q}$, but this is irrelevant), and if t is a root of $C_k(x)$ of multiplicity m , then σt is also a root of $C_k(x)$ of multiplicity m . (Probably all roots of $C_k(x)$ have multiplicity one; see the conjecture below.)

Set $D = \prod_{j=1}^l (2b_j)^{-1}$. By the Lemma D^2 is a rational number, so $\sigma D = \pm D$. Applying σ to (3) yields (since A_n is rational)

$$A_n = \sigma A_n = \pm D \sum_S (-1)^{|S|} (\sigma c_S)^{n+1}, \quad nk \text{ even}. \quad (10)$$

Suppose $t = c_S$, so that m is equal to the number of T for which $c_S = c_T$. Since $c_j > 0$

and $\bar{c}_j < 0$, it follows that $c_S > 0$ if and only if $|\bar{S}|$ is even, and hence $(-1)^{|\bar{S}|} = (-1)^{|\bar{T}|}$ whenever $c_S = c_T$. Thus the coefficient of t^n in (3) when all equal expressions c_X^{n+1} are combined is equal to $(-1)^{|\bar{S}|} Dtm$.

Now all functions $f(n) = \sum_r a_r \gamma_r^n$, where the γ_r 's are distinct nonzero complex numbers and the a_r 's nonzero complex numbers, are different. It follows from (3) and (10) that when k is even the coefficient of $(\sigma t)^n$ in (3) when all equal expressions c_X^{n+1} are combined is equal to $\pm (-1)^{|\bar{S}|} D(\sigma t)m$. Hence exactly m values of T satisfy $\sigma t = c_T$, so that σt is a root of $C_k(x)$ of multiplicity m as desired.

When k is odd (3) is valid only for n even. The above argument applied to $A'_n = A_{2n}$ shows that c_S^2 and σc_S^2 are roots of $C_k(\sqrt{x})$ of the same multiplicity, so that $\pm c_S$ and $\sigma(\pm c_S)$ are roots of $R_k(x)$ of the same multiplicity, completing the proof of (5).

Clearly the numbers c_j are real and, as already observed, satisfy $c_j > 0$, $\bar{c}_j < 0$. From this we immediately have that the roots (or reciprocal roots) of $Q_k(x)$ are real and nonzero, and that exactly half of the roots are positive. A different proof that the denominator of $F_k(x)$, when reduced to lowest terms, has real roots appears in [5, p. 47].

(b) Clearly $c_j > |\bar{c}_j| > 0$, so the largest c_S is uniquely obtained by letting $S = \{1, \dots, l\}$, yielding (6). This largest reciprocal root μ_k cannot be a reciprocal root of $P_k(x)$ since the term μ_k^n appears in (3) with nonzero coefficient, so that μ_k must be a reciprocal root of the *least* denominator of $F_k(x)$. A different proof that the largest reciprocal root of the least denominator of $F_k(x)$ has multiplicity one appears in [1, p. 284].

One can compute $\lim_{k \rightarrow \infty} \lambda_k$ directly from (6) by expressing $\lim_{k \rightarrow \infty} \log \mu_k^{2/k}$ in terms of a Riemann integral in a standard way, yielding

$$\log \lambda = \frac{2}{\pi} \int_0^{\pi/2} \log(\cos x + (1 + \cos^2 x)^{1/2}) dx.$$

The above integral is essentially evaluated, e.g., in [4, p. 1216], and is equal to Catalan's constant G . Hence $\lambda = e^{2G/\pi}$.

Alternatively, Kasteleyn [4] and Temperley and Fisher [9] showed that

$$\lim_{\substack{k, n \rightarrow \infty \\ kn \text{ even}}} A_{n,k}^{2/nk} = e^{2G/\pi}.$$

But (always assuming kn is even)

$$\lim_{k, n \rightarrow \infty} A_{n,k}^{2/nk} = \lim_{k \rightarrow \infty} \left(\lim_{n \rightarrow \infty} A_n^{1/n} \right)^{2/k} = \lim_{k \rightarrow \infty} \mu_k^{2/k}$$

and again (6) follows. This computation of λ_k is mentioned in [6, eqn. (7)].

(c) From (3) and (6), the coefficient a_k of μ_k^{n+1} in A_n is given by (9), so (8) follows.

From (9) and the explicit expression (4) for d_k it is clear that $\lim a_k^{2/k} = (1 + \sqrt{2})^{-1}$. (It is also possible to prove this result without explicitly evaluating d_k , by express-

ing $\lim (2/k) \log (b_1 \cdots b_l)$ as a Riemann integral.)

(d) It follows from the form (3) of A_n and basic facts about rational generating functions [8, Theorem 4.1] that $p_k < q_k$. Then by [7, Proposition 5.2], we have that $q_k - p_k$ is equal to the largest integer m for which $A_{-1} = A_{-2} = \cdots = A_{-m+1} = 0$, where A_{-n} is defined by substituting $-n$ for n in (2) or (3). Clearly by (2) we have $A_{-1} = 0$. On the other hand, since $c_j \bar{c}_j = -1$, it follows that $A_{-2} = \pm A_0 = \pm 1$, and the proof follows.

(e) Since $c_j \bar{c}_j = -1$, we have $c_S \bar{c}_S = (-1)^l$. Hence if k is odd then the reciprocal roots $\pm c_S$ of $Q_k(x)$ come in groups of four of the form $c_S, -c_S, c_S = \pm c_S^{-1}, -c_S = \mp c_S^{-1}$. This implies $Q_k(x) = x^{q_k} Q_k(1/x)$ and $Q_k(x) = Q_k(-x)$. If k is divisible by 4, then the reciprocal roots come in pairs c_S and $c_S = c_S^{-1}$, which implies $Q_k(x) = x^{q_k} Q_k(1/x)$. If $k \equiv 2 \pmod{4}$, then the reciprocal roots come in pairs c_S and $c_S = -c_S^{-1}$, which implies $Q_k(x) = -x^{q_k} Q_k(1/x)$.

Now define

$$\bar{F}_k(x) = \sum_{n>0} A_{-n} x^n.$$

A result of Popoviciu (see e.g. [7, Proposition 5.2]) implies that

$$F_k(x) = -\bar{F}_k(1/x),$$

as rational functions. From (2) and the equality $c_j \bar{c}_j = -1$ it is clear that

$$A_{-n} = (-1)^{(n-1)l} A_{n-2}, \quad A_{-1} = 0.$$

Hence

$$\bar{F}_k(x) = \begin{cases} x^2 F_k(x), & l \text{ even,} \\ x^2 F_k(-x), & l \text{ odd.} \end{cases}$$

Comparing with (11) yields

$$F_k(x) = \begin{cases} -(1/x^2) F_k(1/x), & l \text{ even,} \\ -(1/x^2) F_k(-1/x), & l \text{ odd.} \end{cases}$$

Comparing this result with what was just proved for $Q_k(x)$ (and using $q_k - p_k = 2$) yields the desired properties of $P_k(x)$.

(f) Let $Q(x) = \sum_{i=0}^s \delta_i \binom{s}{i} x^i$ be any polynomial with negative real roots. I. Newton showed (see e.g. [2, Theorem 51]) that $\delta_i^2 \geq \delta_{i-1} \delta_{i+1}$. (This result is in fact valid for any polynomial with real roots.) Now consider for k odd the polynomial

$$Q_k(\sqrt{x}) = \prod_S (1 - c_S^2 x) = \sum_{i=0}^r \gamma_i \binom{r}{i} (-1)^i x^i.$$

Since c_S is real and nonzero, it follows that $c_S^2 > 0$ and hence each $\gamma_i > 0$. By Newton's result, $\gamma_i^2 \geq \gamma_{i-1} \gamma_{i+1}$. Since each $\gamma_i > 0$, this means $\gamma_i \geq \min\{\gamma_{i-1}, \gamma_{i+1}\}$ so that the γ_i 's are unimodal. This completes the proof.

(g) We omit the proof, which is a rather routine consequence of what we already

have shown.

In conclusion we mention the following conjecture.

Conjecture. The polynomial $Q_k(x)$ has distinct roots.

This conjecture is equivalent to the statement that $2^{[(k+1)/2]}$ is the *least* degree of a linear recurrence relation satisfied by A_n (or equivalently, that $P_k(x)$ and $Q_k(x)$ are relatively prime). To see this, note that c_S^n occurs in (3) with nonzero coefficient, so that c_S must be a reciprocal root of the denominator $R_k(x)$ when $F_k(x)$ is reduced to lowest terms. When k is even, this accounts for all $2^{[(k+1)/2]}$ roots of $Q_k(x)$. When k is odd, this only accounts for half the roots of $Q_k(x)$. However, in this case $A_n = 0$ when n is odd. Thus if A_n satisfies (1), then it also satisfies (1) when every term $\alpha_i A_{n+q-i}$ with i odd is deleted. This means that the unique recurrence (1) of minimal degree satisfies $\alpha_{2i+1} = 0$, so $R_k(x) = R_k(-x)$. Hence not only must all the numbers c_S be roots of $R_k(x)$, but also their negatives $-c_S$, and we have again accounted for all $2^{[(k+1)/2]}$ roots of $Q_k(x)$.

Let us point out that although we are unable to decide whether the roots of $Q_k(x)$ are distinct, it is evident from (3) that the least denominator of $F_k(x)$ has distinct roots (because the coefficient of each c_S^n is a constant, rather than a polynomial in n of degree ≥ 1). This answers a question raised in [5, p. 47].

A stronger assertion than the distinctness of the roots of $Q_k(x)$ is the statement that $Q_k(x)$ is irreducible over the rationals. In this regard, J. Lagarias has pointed out to me that the reducibility of $Q_6(x)$ implies the reducibility of $Q_k(x)$ when $k+1$ is divisible by 7. Moreover, Lagarias has proved that $Q_k(x)$ is irreducible whenever $k+1$ is an odd prime $\neq 7$. Hence in this case the above conjecture is valid.

References

- [1] J.H. Ahrens, Paving the chess board, *J. Combin. Theory (A)* 31 (1981) 277–288.
- [2] G.H. Hardy, J.E. Littlewood, and G. Polya, *Inequalities* (Cambridge Univ. Press, Cambridge, 1959).
- [3] J.L. Hock and R.B. McQuistan, A note on the occupational degeneracy for dimers on a saturated two-dimensional lattice space, *Discrete Applied Math.* 8 (1984) 101–104.
- [4] P.W. Kasteleyn, The statistics of dimers on a lattice, *Physica* 27 (1961) 1209–1225.
- [5] D. Klarner and J. Pollack, Domino tilings of rectangles with fixed width, *Discrete Math.* 32 (1980) 45–52.
- [6] E.H. Lieb, Solution of the dimer problem by the transfer matrix method, *J. Math. Phys.* 8 (1967) 2339–2341.
- [7] R.P. Stanley, Combinatorial reciprocity theorems, *Advances in Math.* 14 (1974) 194–253.
- [8] R.P. Stanley, Generating functions, in: G.-C. Rota, ed., *Studies in Combinatorics* (Math. Assoc. Amer., 1978) 100–141.
- [9] H.N.V. Temperley and M.E. Fisher, Dimer problem in statistical mechanics – An exact result, *Phil. Mag.* 6 (1961) 1061–1063.